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Adaptive Calibration Algorithms
for Tunable Analog Peripherals

by Khalil M'hirsi

Master Thesis

Approved by the Examining Committee:

Prof. Mathieu Salzmann
Thesis Advisor

Richard Lengagne
External Expert

Christophe Constantin
Thesis Supervisor

EPFL CVLAB
BC 309 (Bâtiment BC)
Station 14
CH-1015 Lausanne

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*“I struggled for a long time
with surviving.
And no matter what,
you keep finding something
to fight for.”
— The Last of Us*

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Khalil M’hirsi

Abstract

Analog peripherals with continuous depth-sensing hardware offer sub-millimetre actuation precision, yet their firmware often relies on static thresholds that ignore per-user biomechanical variance and degrade under neuromotor fatigue. This thesis addresses that gap by designing an end-to-end click-sensitivity recommendation pipeline for Haptic Inductive Trigger System (HITS) devices, using telemetry streamed from the mouse in order to recommend personalized Actuation Point (AP) and Rapid Trigger (RT) settings. The system makes three contributions: (1) a supervised labeling workflow that first generates frame-level labels from algorithmic criteria (depth geometry, speed, and acceleration thresholds) and then iteratively improves those labels through model-assisted refinement; (2) a four-class temporal intent-detection model—supporting Random Forest, Gradient Boosted Trees, and a dilated residual fully-convolutional network—that combines digital signal preprocessing, class-imbalance-aware sample selection, and context-rich features to identify make-onset, click-hold, and break-onset phases from high-frequency telemetry; and (3) a joint CMA-ES optimizer that simulates the firmware click state machine frame-by-frame and finds Pareto-optimal (AP, RT) configurations that reduce actuation latency while maximizing click reliability through explicit constraints on missed clicks and unintended actuations. Together, these components form a practical recommendation workflow from raw HITS telemetry to interpretable parameter recommendations for click sensitivity tuning.

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1. Introduction

Human-Computer Interaction (HCI) in high-performance computing and competitive esports places stringent demands on input peripherals. In environments where users execute several hundred Actions Per Minute (APM), actuation latency and threshold stability can affect both task performance and musculoskeletal load [23, 32].

1.1 The Evolution of Continuous Sensing Hardware

Peripheral hardware has shifted from binary mechanical switches toward continuous sensing interfaces. In HCI, threshold selection and activation logic have long been recognized as key determinants of usability and speed [18, 21]. Recent Hall-effect and inductive devices expose depth-dependent actuation and reset behavior to firmware, including Rapid Trigger mechanisms for dynamic re-actuation [7, 27, 37]. Commercial systems such as HITS further combine inductive sensing with configurable haptic feedback [10, 19, 20, 31]. Collectively, these developments increase control resolution but also increase dependence on well-calibrated software thresholds [3, 34, 35].

1.2 The Limitations of Static Actuation Thresholds

Despite these hardware leaps, the software interpreting this data remains limited. Current analog peripherals rely on static firmware thresholds that are agnostic to human biomechanical variance. Players exhibit unique control signatures, including micro-hesitations and varying velocity curves, which often clash with rigid hardware logic [18]. When static thresholds are set too aggressively, they fail to filter out physiological noise; when set too conservatively, they introduce unnecessary mechanical pre-travel. This friction results in sub-optimal latency or a high frequency of unintended actuations during high-stress scenarios [21, 34].

1.3 Neuromotor Fatigue and Performance Decay

These hardware trends also intersect with user health. Higher APM in esports has been associated with repetitive strain injuries and tendinopathies

[8, 15, 23]. Biomechanical studies report distinct fatigue patterns for keyboard and mouse interaction [17], and repetitive clicking can degrade aiming accuracy while increasing forearm muscle load [9]. High-APM genres are linked to elevated kinematic demand on wrist extensors [32], with measurable neuromotor adaptation during fatiguing tasks [13]. These findings motivate adaptive calibration that maintains responsiveness while reducing unintended actuation under fatigue.

1.4 Toward Adaptive Intent Detection

One longer-term direction is intent inference before full mechanical actuation. Prior work on surface electromyography (sEMG) and embedded TinyML suggests this is technically plausible [2, 16, 29, 36]. However, this thesis focuses on a narrower and directly evaluable scope: offline calibration from depth telemetry already produced by the peripheral. On-device inference is treated as future work (Section 7.3). Productization and UX integration (including the UX Exploration v2 flow and eventual GHub/GHabits integration) are likewise positioned as future work.

1.5 Thesis and Contributions

The central hypothesis of this thesis is that per-user variation in click telemetry is structured enough to support automatic labeling, statistical modeling, and parameter optimization beyond static default settings. The contributions are:

- A hybrid labeling pipeline for analog HID click intent that derives initial frame-level MAKE_ONSET, CLICK_HOLD, and BREAK_ONSET annotations from algorithmic criteria (geometry, speed, and acceleration) and then improves boundary quality through model-assisted iteration.
- A four-class temporal sequence classifier with two model paths: a shallow path (Random Forest, Gradient Boosted Trees) operating on hand-crafted sliding-window features for sub-second per-session training, and a deep path (dilated residual FCN, DETR-style decoder) operating on raw telemetry with a ± 1530 ms learned receptive field.
- A joint CMA-ES optimizer that simulates the pe-

ipheral firmware’s Rapid Trigger state machine and finds Pareto-optimal (AP, RT) configurations, replacing manual threshold tuning with a data-driven trade-off curve.

While not a primary algorithmic contribution, the work also includes a small HCI-facing component: minigames that replicate the most popular games that provide an environment for user to quickly get a recommendation. The recommendation outputs are structured for interpretability (trade-off views and comparative candidate points), so users can inspect why a suggested AP/RT point is proposed.

2. Background

To understand the design and implementation of an adaptive actuation system, it is necessary to establish the technical foundations of continuous sensing hardware, the mathematical logic of modern trigger parameters, and the biomechanical constraints of the human hand.

2.1 Continuous Depth-Sensing Mechanisms

Traditional mechanical switches operate on a binary principle: a circuit is either open or closed based on a physical contact point. In contrast, continuous depth-sensing switches measure the absolute position of the stem throughout its entire travel distance.

2.1.1 Hall-Effect and Magnetic Sensing. Hall-effect sensors utilize a permanent magnet attached to the switch stem and a sensor on the Printed Circuit Board (PCB). As the magnet approaches the sensor, the Hall voltage varies proportionally to the magnetic field strength [27, 37]. This allows the firmware to map voltage levels to precise displacement values in millimeters. Compared to classic contact switches, this technology reduces dependence on debounce filtering and enables lower-latency signal processing.

2.1.2 Haptic Inductive Trigger System (HITS). Refined in the 2026 Logitech SUPERSTRIKE series, the Haptic Inductive Trigger System (HITS) replaces magnets with an inductive coil architecture [20, 31]. By measuring changes in electromagnetic in-

duction caused by the proximity of a metallic puck within the trigger, HITS provides higher resolution and lower susceptibility to external magnetic interference compared to Hall-effect sensors. Furthermore, HITS integrates localized haptic motors to simulate tactile "bumps" or "clicks" at software-defined depths, decoupling the physical feel of the switch from its electrical actuation point [10].

2.2 Control Parameters: AP and RT

The transition to continuous sensing introduced two primary variables that define the "feel" and speed of an input device.

2.2.1 Actuation Point (AP). The Actuation Point (AP) is the specific depth (d) at which a button-press event is registered. In traditional switches, this is fixed (e.g., 2.0mm). In analog systems, AP is a variable $d_{act} \in [0.1, 4.0]$ mm. While these parameters are characteristically expressed as spatial displacement for keyboards, in the context of mice, a specific actuation depth inherently corresponds to a specific physical force applied onto the keyplates. Lowering d_{act} reduces mechanical pre-travel (or actuation force), allowing for faster response times, but increases the risk of accidental actuations caused by resting finger weight.

2.2.2 Rapid Trigger (RT). Rapid Trigger (RT) eliminates the fixed reset point found in mechanical switches. In a standard switch, the button must travel back up past a specific reset point before it can be pressed again. RT logic instead monitors the direction of travel. As soon as the stem moves upward by a specific threshold (Δ_{up}), the input resets. Conversely, as soon as it moves back down by Δ_{down} , it reactuates [7, 37]. RT mechanisms differ primarily in their reference boundaries: dynamic RT algorithms track the deepest and highest positions continuously along the entire travel path, only terminating this dynamic tracking when the displacement reaches absolute zero (a complete release). By contrast, other RT implementations restrict this dynamic tracking exclusively to the domain below the Actuation Point (AP), instantly terminating the rapid trigger state if the stem ascends past the initial AP threshold. Both methods allow for rapid repeated clicking, which is critical in high-

APM genres.

2.3 Biomechanical Variance and Neuromotor Fatigue

The efficiency of AP and RT is limited by both hardware imperfections and the physiological state of the user. Before examining human motor control, it is important to acknowledge that the raw sensor data is inherently noisy due to analog-to-digital converter (ADC) polling jitter, quantization noise, and non-linearities at the physical extremes of switch travel (top-out and bottom-out crush). However, even assuming a perfectly linear sensor, human motor control is subject to "signal-dependent noise," where the variance of movement increases with the force or speed of the action.

2.3.1 Resting Pressure and Tremors. Even at rest, users exert varying degrees of force on a peripheral. Factors such as cognitive stress or caffeine intake can cause micro-tremors-high-frequency, low-amplitude oscillations in displacement telemetry [14]. If the AP is set to 0.1mm and the user's tremor amplitude is 0.15mm, the system will register "ghost" inputs.

2.3.2 Neuromotor Fatigue Signatures. Extended sessions lead to fatigue in the *flexor digitorum superficialis* and *extensor digitorum* muscles [9, 17]. Fatigue manifests in two measurable telemetry changes:

1. **Velocity Decay:** The "snap-back" velocity (the speed at which a finger releases a key) decreases as the extensor muscles tire [33].
2. **Resting Weight Shift:** As the flexor muscles lose tonicity, the resting weight of the finger often increases, leading to deeper "resting" displacement in the analog sensor [13, 15].

2.4 Constraints of Offline Calibration (Scope Context)

This thesis focuses on offline calibration from depth telemetry. While continuous telemetry at high polling rates could theoretically be processed via on-device inference using quantized INT8 models on peripheral MCUs [16], hardware-in-the-loop deployment requires firmware-level access

and strict latency budgets [24]. Consequently, on-device Edge ML is treated as future work, and all calibration pipelines in this study operate offline.

2.5 Signal Processing and Optimisation Foundations

The adaptive calibration pipeline draws on several established signal processing and machine learning methods. This section describes their mathematical properties; their problem-specific justifications appear in the Design chapter.

2.5.1 Savitzky-Golay Smoothing. Savitzky-Golay (SG) filtering fits a polynomial of degree p to each sliding window of W samples in a least-squares sense and evaluates the polynomial at the window centre [28]. Because the convolution coefficients are precomputed from the normal equations, the runtime cost is $O(n)$ regardless of window width, identical to a moving average. Crucially, the polynomial fit exactly reproduces moments up to degree p , so peak positions and edge inflection points in the signal are not distorted—a property absent from symmetric Gaussian or box filters, which both shift extrema toward the window centre.

2.5.2 Hermite Cubic Interpolation: Akima and Makima. Akima interpolation constructs a piecewise cubic Hermite spline whose tangent at each knot is a weighted average of the four adjacent finite differences, with weights inversely proportional to the absolute difference between consecutive slopes [1]. This weighting isolates the tangent estimate from outlier slopes: a single anomalous interval does not propagate ringing across the spline, unlike a natural cubic spline whose tangents are coupled globally. Akima does not enforce monotonicity, so it can represent non-monotone smooth curves. Modified Akima (Makima) adds a half-mean-slope regularisation term to the weight formula, preventing zero-weight degeneracy when two adjacent slopes are equal and opposite—a degenerate case that arises frequently in quantized sensor signals.

2.5.3 Covariance Matrix Adaptation Evolution Strategy. To optimize physical hardware thresholds like AP and RT, we require black-box optimizers that

handle continuous, correlated variables well. CMA-ES is a derivative-free evolutionary optimiser for continuous search spaces [12]. It maintains a multivariate Gaussian distribution over candidate solutions and, at each generation, moves the distribution mean toward the best-performing candidates and adapts the full covariance matrix to reflect the local curvature of the objective landscape. This covariance adaptation lets the algorithm exploit correlations between parameters (such as the interdependent relationship between optimum AP and RT points) and navigate elongated, ill-conditioned valleys that cause gradient-free methods with diagonal step sizes (e.g. Nelder-Mead or coordinate descent) to stall. CMA-ES converges in $O(n^2)$ function evaluations for an n -dimensional problem, far fewer than the $O(N^n)$ evaluations required by grid search over a width- N grid.

2.5.4 Ensemble Methods: Random Forests and Gradient Boosting. For the offline ML pipeline, ensemble methods are particularly suited for extracting features from sliding-window time-series data because they are robust to scale differences and irrelevant features. A Random Forest [5] builds an ensemble of decision trees, each trained on a bootstrapped subsample with a random subset of features considered at each split, and averages their posterior class probabilities. The double randomisation decorrelates tree errors, yielding ensemble variance lower than any individual tree without significantly increasing bias. Gradient Boosted Trees [11] instead fit trees sequentially: each tree is trained to correct the pseudo-residual of the current ensemble via gradient descent on a differentiable loss. This staged correction reduces bias relative to a Random Forest, at the cost of longer training time and greater sensitivity to hyperparameter tuning.

2.5.5 Deep Temporal Models: FCN and DETR-style Architectures. While ensemble methods excel with hand-crafted features, deep learning models can automatically extract hierarchical representations directly from raw, unaligned telemetry data. Fully Convolutional Networks (FCNs) adapted for time-series employ stacked layers of 1D dilated convolutions. Dilated convolutions expand the temporal re-

ceptive field exponentially without requiring pooling layers that destroy temporal resolution, making them well-suited for capturing long-range dependencies in continuous analog sensor streams [4].

To transform these continuous feature maps into discrete temporal boundaries (such as click onsets or releases), transformer-based architectures modelled after the DETECTION TRANSFORMER (DETR) [6] can be adapted for 1D sequences. A DETR-style decoder utilizes multi-head cross-attention to match a fixed set of learned "event queries" against the FCN's sequence features. This frames sequence labeling as a direct set prediction problem, allowing for the precise localisation of temporal boundaries and bypassing heuristic post-processing tools like non-maximum suppression.

3. Related Work

This chapter surveys prior work across the five domains that the proposed system draws on and contributes to: adaptive threshold methods for HID devices, temporal sequence labeling from biosignals, programmatic labeling and weak supervision, dilated convolutional architectures for sequence modeling, and transformer-based event decoding for temporal sequences.

3.1 Adaptive Threshold Methods for HID Devices

The problem of optimizing actuation parameters for human input devices has a long history in HCI, though prior approaches have relied on fixed heuristics rather than per-user learned models.

Kim et al. [18] introduced *impact activation*, an alternative actuation scheme for mouse buttons in which a click is registered at the first contact force peak following a rapid finger descent, rather than at a fixed displacement threshold. Their evaluation showed faster reaction times and fewer double-click errors under time pressure. Importantly, impact activation still relies on a single fixed heuristic (the first force peak), making it unsuitable for users with atypical contact dynamics, and it does not adapt to within-session fatigue. This thesis learns thresholds from the user's own recorded telemetry rather than applying a globally fixed policy.

Trewin [34] proposed the *Dynamic Keyboard*,

which automatically adjusts keyboard response parameters (minimum press duration, maximum bounce interval) to accommodate users with motor impairments, by observing an initial calibration sequence and adjusting parameters using rule-based adaptation. The Dynamic Keyboard targets binary on/off switch events and adjusts temporal debounce windows; it does not exploit continuous analog depth signals, and its adaptation rules are domain-expert heuristics rather than models trained from user data. The proposed system is complementary in motivation—both aim to reduce unintended actuations—but operates at a different signal level (continuous depth at 1 kHz vs. binary event timing) and uses a data-driven model rather than hand-coded rules.

To the best of our knowledge, peer-reviewed literature has not yet reported a complete learning-based pipeline for automatic joint tuning of Rapid Trigger (RT) and Actuation Point (AP). Current commercial implementations [19, 37] primarily expose RT and AP as user-adjusted controls without closed-loop personalization. The joint CMA-ES optimizer in this thesis therefore positions RT/AP tuning as an explicitly optimized per-user objective rather than a manual configuration task.

3.2 Temporal Sequence Labeling in Biosignal Processing

Detecting click intent from continuous depth telemetry is an instance of temporal sequence labeling, a problem that has received substantial attention in the context of richer biosignals.

Surface electromyography (sEMG) is the closest related modality. Wang et al. [36] presented ReactEMG, which uses masked autoencoder pre-training on sEMG to achieve stable, low-latency intent detection from eight-channel surface electrode recordings. Antuvan and Masia [2] applied Linear Discriminant Analysis to simultaneous classification of wrist and finger movements from sEMG, demonstrating real-time throughput on embedded hardware. Both systems operate on multi-channel electrode signals that encode rich neuromuscular information before mechanical actuation occurs; their signal-to-noise advantage over single-sensor

depth telemetry is substantial. The proposed system, by contrast, uses only the depth signal already present in the peripheral itself, requiring no additional sensing hardware. This imposes a harder classification problem—depth telemetry cannot observe pre-actuation motor intent—but removes the deployment barrier of wearable electrodes.

Keystroke dynamics literature [22, 30] models typing patterns for biometric authentication from binary key event timestamps. TypeFormer [30] applies a Transformer architecture to sequences of inter-key timings and achieves high verification accuracy on mobile devices. These methods treat the keystroke as an atomic event and model the temporal pattern between events; they do not model sub-millisecond intra-click dynamics. The proposed system operates at a finer granularity—frame-level classification at 1 kHz—which requires a different feature engineering approach and exposes a different class of failure modes (contact-bounce artefacts, quantization noise) absent from binary event streams.

3.3 Weak Supervision and Algorithmic Bootstrapping

The geometric auto-labeler introduced in Section ?? fundamentally differs from programmatic weak supervision frameworks such as Snorkel [26]. Snorkel defines *labeling functions*—heuristic rules, distant supervision signals, or domain-expert logic—that each assign noisy labels to unlabeled examples, and combines them via a generative model that learns the accuracy and correlations of each labeling function. The combined label distribution is then used as the training signal for a downstream discriminative model.

Instead of deploying the geometric auto-labeler as an automated weak supervision signal for model training, the proposed system employs it strictly as an algorithmic bootstrapping assistant for human-in-the-loop annotation. Purely algorithmic segments (i.e., those assigned by the auto-labeler but not verified) are explicitly stripped from the dataset during the training pipeline; the machine learning architectures are trained exclusively on user-verified and manually adjusted boundaries.

This conservative data policy ensures high-fidelity ground truth and prevents the model from merely mirroring the systematic biases and plateau ambiguities of the underlying geometric heuristic. As a result, the proposed system acts as a strongly supervised learning pipeline accelerated by algorithmic pre-labeling, rather than a weakly supervised system relying on noisy heuristics. To account for natural human inconsistency in selecting the exact millisecond of an onset during this manual process, the system employs asymmetric soft labels (described in Section ??) that encode physiological directional uncertainty into the verified targets, rather than treating boundary annotations as rigid, perfect temporal observations.

3.4 Dilated Convolutions for Sequence Modeling

The OfflineFCN architecture adapts the dilated causal convolution structure introduced by van den Oord et al. in WaveNet [25] for raw audio generation. WaveNet stacks dilated causal convolutional layers with exponentially increasing dilation rates to achieve a large receptive field with manageable parameter counts. The dilation schedule $d_i = 2^i$ allows the stack to cover $\sum 2^i$ steps of context with $O(\log N)$ layers rather than the $O(N)$ layers that a fixed-dilation architecture would require.

The OfflineFCN departs from WaveNet in one critical dimension: it uses symmetric (non-causal) padding rather than causal (left-only) padding. This doubles the effective receptive field per layer for the same kernel size, since context is drawn from both directions—enabling the ± 1530 ms receptive field of Equation (??) to be achieved with only 8 blocks. The trade-off is that the resulting model cannot be deployed for streaming inference, since each output depends on future samples. This is appropriate for the offline analysis mode but excludes the model from the on-device path described as future work.

3.5 Transformer Decoders for Temporal Event Prediction

Recent sequence models increasingly frame boundary detection as an event-set prediction problem rather than strictly per-frame labeling. Transformer

decoders with learned queries have shown that global context and query interaction can model sparse event structure without hand-crafted transition constraints. In this framing, each query predicts an event hypothesis and confidence, and training resolves assignment via bipartite matching so duplicate detections are penalized directly.

For click-intent segmentation, this perspective is relevant because make/break boundaries are sparse relative to background frames and are naturally interpreted as events on a timeline. A DETR-style decoder [6] on top of a high-context temporal backbone can therefore replace rule-based post-processing or transition-constrained decoders with a single learned event predictor. The trade-off is computational: query-based decoding is heavier than shallow per-frame classifiers and demands stronger regularization on small per-user datasets.

4. Design

This chapter details the architecture and algorithmic design of the proposed adaptive actuation system.

4.1 System Architecture

The system utilizes a multi-layered software and hardware stack designed to bridge high-frequency physical telemetry with a dynamic optimization engine. At the hardware layer, a modified PRO X2 Superstrike mouse streams continuous 120-level analog depth values at 1 kHz. This bypasses typical firmware constraints that generally limit reporting to mere 4-bit precision at 65 Hz and block reading after haptic events. The hardware communicates with the host operating system via GXT-Devio, an OS-level bridge that interfaces high-level prototyping environments (TypeScript/JavaScript) with production-grade HIDPP protocols. Sitting above this is Nightfall, a background sensor service that orchestrates and synchronizes the analog telemetry with other biometrics and OS events. Finally, GSense serves as the graphical orchestration client, providing the user dashboard and managing the downstream machine learning pipelines.

4.2 Data Collection and Labeling

Raw analog signals are inherently noisy. Before labeling, the depth telemetry is processed using Modified Akima (MAKIMA) interpolation. MAKIMA was specifically selected because it preserves signal monotonicity and prevents the severe overshoots characteristic of standard cubic splines.

To alleviate the bottleneck of manual annotation, the system employs an automated mathematical labeling pipeline to bootstrap the data:

- **Make Intention:** Detected at the first non-zero analog depth value marking the start of a downward press.
- **Break Intention:** Detected at the exact frame where the depth speed reverses (where local velocity crosses zero), indicating the start of a release.

While this algorithmic bootstrapping rapidly processes the bulk telemetry, training a high-fidelity machine learning model demands human-verified ground truth. Consequently, a curated subset of these sessions underwent rigorous manual annotation. The human labeling mindset focused strictly on physical, biomechanical intent—visually distinguishing and discarding involuntary micro-tremors and resting-weight drift, while preserving genuine, deliberate presses regardless of whether they resulted in a successful in-game action. In total, [X] hours of telemetry, purposely curated to capture high biomechanical variance across genres, were processed. This yielded a highly controlled training baseline of approximately 30,000 fully verified click cycles (broadly distributed across MOBA and tactical FPS environments).

Because 80–90% of a user’s session consists of idle background noise, an entropy scoring system based on statistical variance is applied. This mechanism trims idle segments from the dataset, ensuring the downstream models train efficiently and do not become destructively biased toward the background class.

4.3 Click-Intent Sequence Modeling

To understand user intent, the system models the data as a supervised time-series segmentation

problem, mapped down to a per-frame classification task. The core feature engineering relies heavily on extracting analytical "distant context probes" that provide models with past and future temporal context, combined with instantaneous features like analog speed, digital signals, and distance to dynamic thresholds.

The primary algorithm used for inference is a Random Forest (RF) classifier. More advanced architectures, such as Gradient Boosted Trees (GBT) accommodating custom loss functions, and Fully Dilated Convolutional Networks (FDCN) for direct raw sequence temporal modeling, operate within the same structural framework.

4.4 Parameter Optimization

Once intent boundaries are parameterized, the system tests potential hardware settings without physically altering the peripheral. The engine simulates the analog mouse click Finite State Machine (FSM) frame-by-frame, testing its actuation behavior both with and without Rapid Trigger logic enabled.

The optimizer relies on the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) to explore the parameter space. Candidates are evaluated through a multi-objective hand-crafted loss function that balances Recall (not missing clicks), Precision (avoiding accidental misclicks), and Latency (an early prediction penalty aimed at minimizing the temporal distance between simulated triggers and true user intent).

5. Implementation

This chapter describes the software and hardware implementation of the system designed in Chapter 4.

5.1 Frontend and Telemetry Stack

The user-facing layer is built within GSense, providing the orchestration dashboard, recording states, and test minigames. Telemetry ingestion is managed primarily by GXT-Devio via HIDPP and seamlessly synchronized by the Nightfall background service. This layered composition ensures that 1 kHz hardware polling runs safely out-of-process, leaving the GSense frontend free to organize workflows

without stalling the main analytical execution arrays.

5.2 Machine Learning Backend

The machine learning pipeline is encapsulated in a dedicated Python intent-detection backend. During preprocessing, SciPy-based MAKIMA interpolation and statistical variance filtering are executed to sanitize and score the 1 kHz stream entropy. The Random Forest classifier relies heavily on scikit-learn logic, converting the sliding windows of distance/velocity probes into flattened tabular feature arrays. Experimental sequence-level architectures, specifically the FDCN approaches, are developed using PyTorch.

5.3 Optimization Pipeline Integration

The optimization phase bridges the machine learning framework back into direct hardware behavior suggestions. The CMA-ES simulation implements custom finite state machine logic algorithmically mirroring the precise internal trigger mechanisms for the PRO X2 Superstrike. The frontend integration directly extracts the optimizer's outputs to satisfy critical user-sentiment constraints observed during testing: displaying the resultant AP and RT adjustments not as a simple black-box recommendation, but as graph-supported "educated guesses" illustrating the explicit Recall/Precision tradeoff improvements.

5.4 Data Acquisition Campaign

The development and training of the machine learning backend required the construction of a robust ground-truth dataset spanning diverse ergonomic and biomechanical profiles. Data was collected from 69 participants via active field deployments at two major public gaming events (BIC and PolyLAN).

5.4.1 Controlled Task Dataset (BIC Event). The first deployment cohort ($N = 26$) comprised a mix of highly ranked competitive gamers and casual novices playing *Counter-Strike 2* (CS2). The protocol ran for approximately 60 minutes per user and was strictly structured to isolate specific click dynamics:

- **Configuration A/B Testing:** Following a uniform warmup phase, users played two 15-minute game blocks. The device hardware profile was hot-swapped halfway through the session, alternating between an aggressive configuration (AP at 1, RT at 1) and a highly conservative configuration (AP at 5, zero Rapid Trigger). The presentation order was randomized to mitigate sequence bias.
- **Weapon Rotations:** Within each 15-minute block, users were instructed to rotate their active weapon every five minutes. The arsenal was specifically chosen to evoke distinct motor patterns: the *USPS* pistol for rapid individual tapping fatigue, the *M4A1* rifle for sustained tracking and spraying, and the *AWP* sniper rifle for slow, deliberate holding and release.

Qualitative user feedback regarding the switch feel was collected asynchronously after the sessions via standardized surveying.

5.4.2 Free-Play Dataset (PolyLAN Event). The secondary dataset ($N = 43$) focused on unstructured free-play in a high-density LAN environment. Participants engaged in approximately 40-minute play sessions utilizing their preferred competitive title (*League of Legends*, *CS2*, or *Valorant*). Similar to the controlled dataset, the hardware configurations were silently swapped from conservative to aggressive baseline profiles near the 20-minute mark to capture a wider net of organic responses.

The aggregated dataset generated from these two campaigns ultimately produced hundreds of hours of raw 1 kHz telemetry, demanding the variance entropy scoring (detailed in Section 4) to isolate periods of genuine engagement before entering the classification pipelines.

6. Evaluation

This chapter evaluates the performance of the machine learning classifiers and the downstream hardware parameter optimizer.

6.1 Evaluation Setup

6.1.1 Metrics. The segmenting classifier's quality is primarily measured via per-class classification ac-

curacy alongside the absolute temporal detection error, expressed as the average millisecond divergence from the ground truth intention timestamp. The parameter optimizer is evaluated on its success in minimizing the simulation's tripartite loss: maintaining peak precision and recall whilst simultaneously minimizing latency.

6.1.2 Baselines. The optimizer evaluates hardware configuration changes linearly against the fixed threshold baselines (e.g., default firmware behaviour). The underlying detection engine evaluates success compared to raw heuristics mapping logic.

6.2 Click-Intent Classifier Results

Tasked with interpreting the analog stream, the Random Forest model driven by "distant context probes" achieved extraordinary performance levels. It successfully attained 97% segmentation accuracy on distinguishing genuine play intent from background data. Furthermore, the model provided extremely tight temporal bounding, showing merely a 7 ms average detection error on labeling constraints.

6.3 Ablation Studies

6.3.1 Variance Entropy Scoring. Considering that 80–90% of raw game session data operates purely as noisy background idleness, attempting to run models without first applying the variance-based entropy score trimming produces severe data imbalance, crippling model convergence and execution speeds.

6.3.2 Contextual Probes. Ablating the "distant context probes" verifies that analyzing isolated instant frames limits classification to naive static thresholding. Including pre- and post-frame analytical slices serves to contextualize the momentum of a single finger, allowing 97% classification precision.

6.4 Parameter Optimizer Results

The CMA-ES optimizer effectively simulated the hardware switch logic, validating the system's ability to hunt specific hardware tolerances to uniquely match physical telemetry. It successfully discovers

configurations that improve objective measures by collapsing latency gaps within the parameters of strong recall and precision.

6.5 End-to-End User Validation Protocol (Project ARROW)

To close the loop from theoretical model accuracy to practical HCI outcomes, a structured end-to-end user validation protocol (Project ARROW) is designed. This evaluation assesses the subjective trust users place in the calibration recommendations and quantifies objective performance changes during actual gameplay.

6.5.1 Study Objectives. The user validation protocol operates on three primary objectives:

1. **Subjective Trust & Perception:** Assessing users' willingness to trust an autonomous calibration tool and accept its recommendation, specifically tracking feedback regarding how the "intent proficiency score" and recommendation UI are interpreted.
2. **Objective Performance (Zapper Trajectory):** Utilizing a custom standalone aiming minigame (Zapper) to analyze "stop-to-click" latency. A measurable reduction in this mechanical threshold latency verifies the optimizer's effectiveness over human-adjusted defaults.
3. **Mislick Monitoring (Performance Loss):** Actively logging any catastrophic regressions, explicitly tracking if a recommended configuration leads to unintended inputs or if the user actively rejects the calibration due to lack of trust.

(Placebo testing was intentionally excluded from the scope due to participant sample size limitations).

6.5.2 Protocol Structure. The 60-minute evaluation protocol is divided into four distinct phases:

- **Phase 1: Pre-Session Onboarding (0–5 mins).** Users define their base ergonomic settings (DPI). The mouse is strictly reset to a conservative standard configuration baseline. Initial "out-of-the-box" impressions are recorded and a participant ID is assigned.

- **Phase 2: Baseline Calibration (5–20 mins).** Utilizing a think-aloud protocol, users engage with the initial calibration wizard and play for 10–15 minutes while telemetry is recorded. This phase concludes with a 1-minute baseline performance round in the Zapper minigame.
- **Phase 3: Post-Optimization UX (20–40 mins).** Users review the UI presenting the optimization recommendation. If a user rejects the algorithm’s configuration due to a lack of trust, it is recorded as a critical UX failure alongside their justification. Users then apply the optimization and play a secondary session, culminating in a final 1-minute Zapper performance test to capture delta-latency improvements.
- **Phase 4: Offboarding & Validation (50–60 mins).** Technical verification ensures the GSense tracking history is correctly encapsulated for the specific participant ID, followed by final qualitative user surveying and debriefing.

6.6 Discussion

6.6.1 Strengths. The solution successfully operationalizes continuous hardware capability with machine learning inference, addressing the known difficulty of correctly tuning analog actuation devices manually. Identifying 1 kHz intent with 7 ms precision empowers the system to drastically override default performance blocks efficiently.

6.6.2 User Trust and Adoption Roadblocks. User evaluation provided a stark insight into adoption: users categorically distrust entirely autonomous hardware modification engines acting as black boxes. For production scale viability (such as integration into GHub and GHabits platforms), visual performance validations alongside objective data transparency (graphs outlining the recommended change) are structurally essential to foster adoption.

7. Conclusion

7.1 Summary

This thesis presented an offline adaptive calibration pipeline for analog peripherals with continuous depth sensing. The work addresses a practical limi-

tation of current firmware: static actuation thresholds that do not adapt to per-user biomechanics, device variability, or within-session fatigue.

The main contributions are:

- A **hybrid labeling pipeline** that derives initial frame-level MAKE_ONSET, CLICK_HOLD, and BREAK_ONSET annotations from algorithmic criteria (depth geometry, speed, and acceleration) and then refines boundary quality through model-assisted iteration. A geometric weak-supervision heuristic is combined with asymmetric soft labels to model directional timing uncertainty at boundaries.
- A **four-class temporal sequence classifier** with two model paths. The shallow path (Random Forest, Gradient Boosted Trees) operates on hand-crafted sliding-window features and supports rapid per-session retraining. The deep path (dilated residual FCN with DETR-style decoding) operates on raw four-channel telemetry with a learned ± 1530 ms receptive field, trading higher context for higher compute cost.
- A **joint CMA-ES optimizer** that simulates the peripheral firmware’s Rapid Trigger state machine frame-by-frame and finds Pareto-optimal (AP, RT) configurations under latency and false-positive objectives, replacing manual threshold sweeps with a data-driven trade-off analysis.

In aggregate, these components provide an end-to-end workflow in which a user recording can be automatically labeled, modeled, and optimized to produce evidence-based AP/RT recommendations with minimal manual intervention.

7.2 Limitations

Several limitations constrain the generality of the current system:

- **Small evaluation cohort.** The system was evaluated on a limited number of users and sessions. Cross-user generalization—whether a model trained on one user’s data transfers meaningfully to another—was not demonstrated at scale. The per-session training approach mitigates this by avoiding cross-user transfer entirely, but the cold-start problem for first-time users

with no prior recordings is not addressed.

- **Auto-labeler timing errors.** The geometric labeling heuristic introduces systematic timing bias of up to W_b frames. While asymmetric soft labels partially absorb this uncertainty, they do not eliminate it. Make-onset F1 is sensitive to the plateau patience parameter, confirming that labeler quality is a first-order determinant of classification performance.
- **Offline-only inference.** The OfflineFCN’s symmetric non-causal padding is incompatible with streaming inference on the peripheral MCU. Any deployment on the on-device path would require a causal (left-padded) architectural variant, which reduces the per-layer receptive field contribution by half.
- **Drift channel unused in the optimizer.** The drift channel ($\text{EMA}_{20} - \text{EMA}_{200}$) is computed and exposed in the session review UI but is not integrated into the optimizer’s objective function. As a result, the system does not currently adapt AP/RT recommendations in response to detected fatigue-induced resting-weight shift.

7.3 Future Work

1. **On-device deployment.** Quantize the OfflineFCN to INT8 and evaluate inference latency on the peripheral RISC-V MCU. Develop a causal (left-padded) variant compatible with streaming inference. A causal model would convert the offline classifier into a real-time intent detector, enabling the pre-report filtering described in the Background.
2. **Multi-session accumulation.** Train on pooled data from a user’s full recording history rather than re-training from scratch each session. A multi-session model accumulates evidence about the user’s long-term biomechanical signature, reducing label noise from individual sessions and improving performance on edge cases (fatigue-modulated dynamics, device warm-up transients).
3. **Drift-aware re-calibration.** Integrate the drift channel into the optimizer’s objective: trigger an automatic re-calibration recommendation when sustained positive drift

exceeds a per-user threshold, indicating flexor-fatigue onset. This closes the loop between the fatigue diagnostic described in Section ?? and the AP/RT adaptation.

4. **Cross-user pre-training.** Pre-train the FCN on pooled data from multiple users and fine-tune per-session, reducing the cold-start problem for new users with short recording histories. A pre-trained backbone should require fewer labeled frames to reach stable performance than a randomly initialized model.
5. **Product and UX integration.** Integrate the recommendation workflow into the UX Exploration v2 flow so users can inspect why a suggested AP/RT point is selected (trade-off plots, confidence signals, and comparable alternatives), then harden the same interface and telemetry contracts for downstream integration into GHub/GHabits.

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